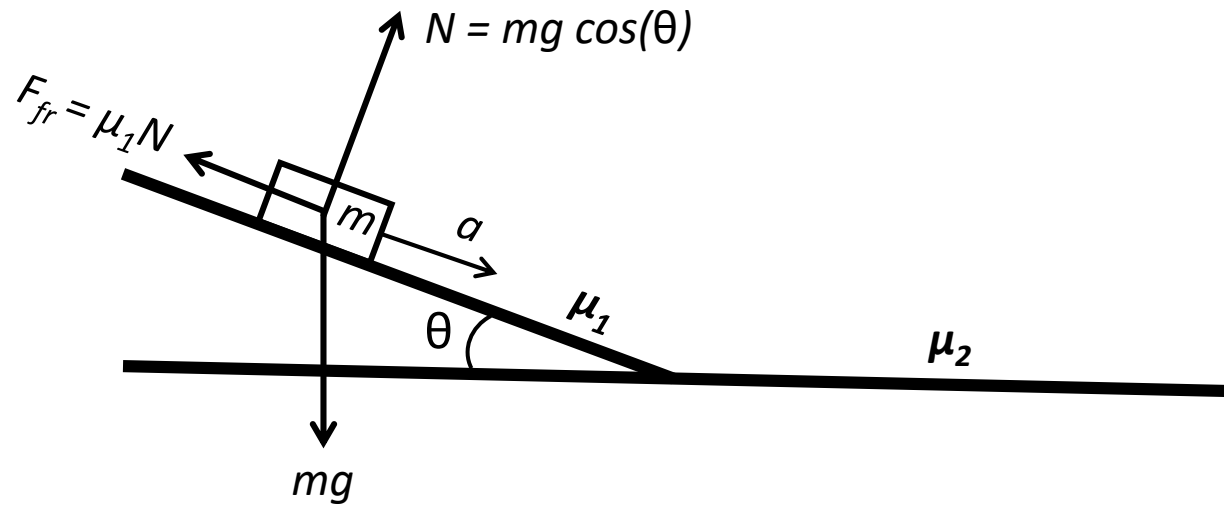


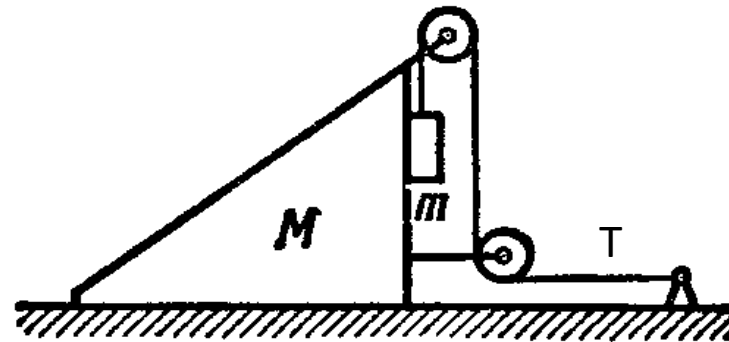
1.

Object Sliding Down an Inclined Plane

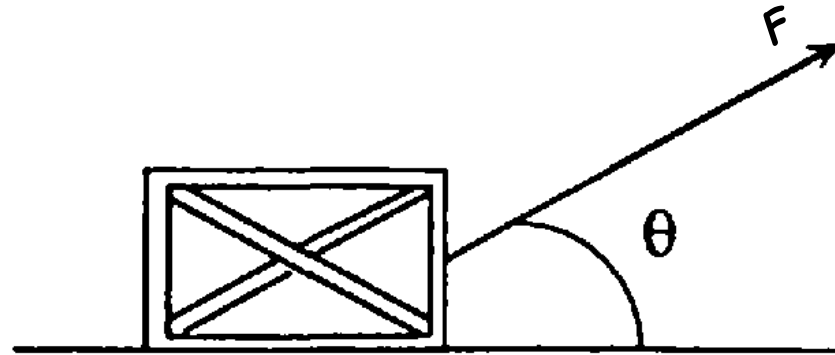
An object of mass $m = 4.0$ kg, starting from rest, slides down an inclined plane of length $l = 3.0$ m. The plane is inclined by an angle of $\theta = 30^\circ$ to the ground. The coefficient of kinetic friction $\mu_k = 0.2$. At the bottom of the plane, the mass slides along a rough surface with a coefficient of kinetic friction $\mu_k = 0.3$ until it comes to rest. The goal of this problem is to find out how far the object slides along the rough surface.



2. The friction coefficient between the wedge (of mass M) and the bar (of mass m) is K . The masses of the thread and the block are negligible. Find the bar acceleration.

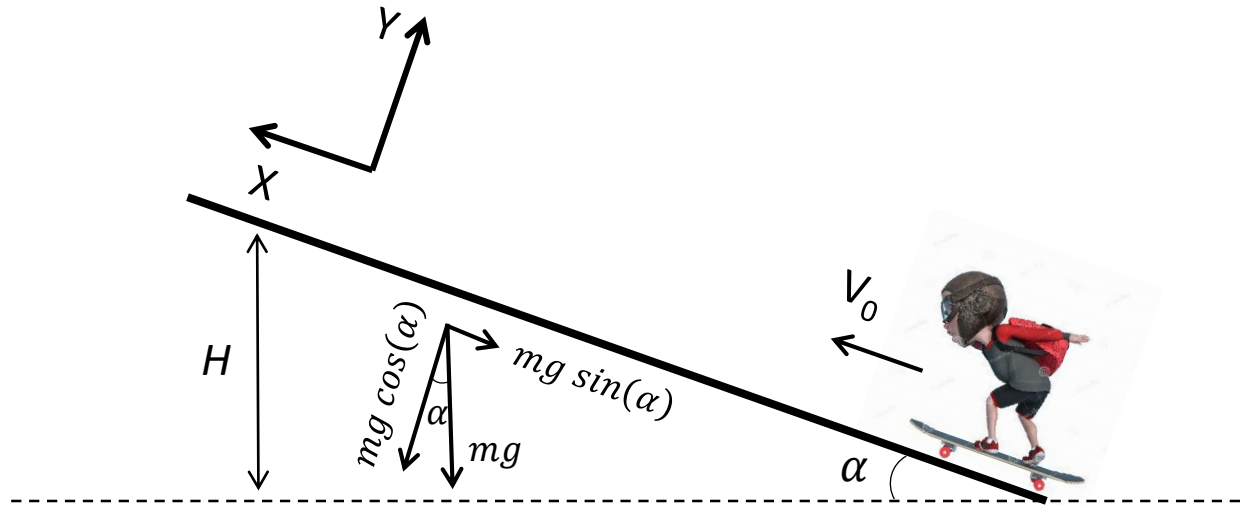


3. A packing crate of unknown mass is pulled across the floor at constant velocity by a cable attached to the crate. The angle θ_0 corresponds to the minimum force to move the crate. What is the friction coefficient between the crate and the floor?

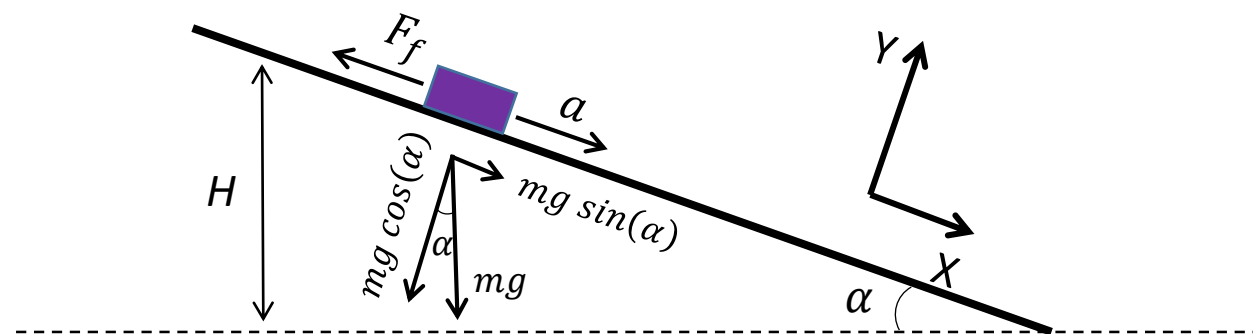


4.

A man rides a skateboard. His initial velocity at the foot of a hill is V_0 . What is the maximum height H he will reach, if the hill makes an angle α with the horizon? The friction coefficient of the skateboard is μ .



5. The bar starts to slide down with a zero initial velocity along the plane with inclination of α with respect to horizon. The friction coefficient varies according to the expression $k(x) = f x$, where f is a positive constant, x is the distance traveled. Find (a) the total distance the bar traveled and (b) its maximum velocity.



6. A plank of mass m_1 with a bar of mass m_2 placed on it lies on a smooth horizontal plane. A horizontal force growing with time t as $F = p t$ (p is constant) is applied to the bar. Find how the accelerations of the plank a_1 and of the bar a_2 depend on t , if the coefficient of friction between the plank and the bar is equal to k .

